

Usable Anonymity in Reproductive Health Privacy

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Abstract—Anonymity is a key strategy for internet users to manage privacy and mitigate data surveillance risks. While tools like Tor support strong anonymity, they are often inaccessible to non-technical users, who instead rely more on platforms’ built-in privacy features to seek anonymity. However, the usability of these features remains underexplored. This study focuses on period and fertility tracking (PFT) apps as a research context due to heightened privacy concerns about reproductive health surveillance, which intensify users’ need for usable privacy designs to achieve anonymity. We conducted usability testing with 32 participants across six PFT apps, revealing that the usability of anonymity-preserving features (APFs) significantly influences users’ privacy perceptions. We identified key usability challenges, such as a lack of transparency regarding APFs and unintuitive processes for avoiding disclosure of personally identifiable information (PII). To address these issues, we propose recommendations for more accessible and usable anonymity in online platforms.

1. Introduction

Anonymity refers to the ability to conceal one’s identity [1], [2], [3]. Seeking anonymity has become a crucial strategy for internet users aiming to manage their online privacy and mitigate risks associated with data surveillance [4], [5]. Security and privacy (S&P) research on anonymity has extensively investigated anonymity tools like Tor, which provide strong protections by masking IP addresses, traffic routes, and user locations [6], [7], [8], [9]. While these tools are highly effective for anonymous browsing or circumventing censorship, they are generally more accessible to tech-savvy individuals [10]. Their underlying technical complexity—such as understanding relays, circuits, or routing—can pose a barrier for everyday users, who may lack the mental models needed to fully adopt or trust such tools [10].

As a result, general users often rely on platforms’ built-in privacy features to pursue anonymity [4]. While prior anonymity-related research in the S&P field tends to empha-

size technical identifiers such as IP addresses or circuit identifiers [7], [11], users often take a broader view—associating anonymity with managing the visibility of a wide range of personally identifiable information (PII)—like names, birthdays, and email addresses—alongside technical identifiers [4], [12]. To manage various forms of PII, users often utilize platforms’ built-in privacy features that support anonymity through various means, such as anonymous and pseudonymous usernames that allow users to create accounts detached from their real identities [13], and anonymous email services that allow usage without disclosing personal details like names or technical identifiers like IP addresses [14].

While previous research explored how users leverage various built-in features for preserving anonymity on online platforms [13], [15], [16], there is no clear consensus on how anonymity should be designed in practice. There remains a significant gap in understanding these features’ usability and how that usability impacts privacy perceptions.

To address these gaps, we investigate the usability of anonymity-preserving features (APFs) to understand users’ expectations of these features and provide insights into how these designs can better support users in achieving their anonymity goals. **Here we define APFs as a set of built-in platform privacy features that enable users to seek, approach, or achieve anonymity.** This definition acknowledges that anonymity is not absolute or binary, but has varying degrees and robustness in real-world situations [1], [3], [17]. Additionally, this framing allows us to evaluate a broad range of features and user experiences, offering a more inclusive and realistic understanding of anonymity in practice.

Acknowledging that the design and interpretation of APFs are highly influenced by context—that is, by the specific risks, norms, and sensitivities of a given setting—we focus our inquiry on period and fertility tracking (PFT) apps. These apps present a particularly relevant context due to heightened privacy concerns about identification and reproductive health surveillance, which amplify users’ need

for APFs. PFT apps (e.g., Clue, Flo) collect intimate health data—including menstrual cycle dates, physical symptoms, and sexual activity—to predict fertility and menstruation patterns, and even infer pregnancy status [18], [19]. Representing a vital segment of the growing “Femtech¹” industry, these apps have gained widespread adoption, with nearly one-third of women in the U.S. reporting their use [23]. However, the collection of sensitive health information—along with PII such as names, email addresses or credit card numbers—raises intensified privacy concerns considering the pervasive reproductive health surveillance landscape [24], [25]. Users fear that their data could be accessed, shared, or weaponized by a range of adversaries, including government agencies, insurers, advertisers, and anti-abortion groups [26], [27], [28], [29]. These concerns reflect a threat landscape in which personal data may be exploited through mechanisms such as third-party tracking, linkage of sensitive health data to user identity, subpoena requests, or commercial data sales [27], [30]. Technical analyses further confirm users’ concerns, revealing opaque cross-jurisdictional data sharing and, in some cases, insecure transmissions vulnerable to interception [31], [32]. To mitigate these risks, users have expressed a strong desire for robust privacy features that support anonymity [26], [29].

In response, some PFT apps—like Flo and Cycles—have publicly claimed to offer privacy features to provide an “anonymous experience” [33], [34]. Grounding our investigation of usable APFs in the critical need for anonymity within the context of PFT apps where concerns about data surveillance are paramount, we pose the following research questions:

RQ1: How usable are APFs in PFT apps?

RQ2: How does the usability of APFs influence user privacy perceptions in PFT apps?

RQ3: What are the main usability challenges users face when interacting with APFs in PFT apps?

To answer these questions, we conducted 32 usability testing sessions on six systematically selected PFT apps (See Section 3.2(3) for more details) with participants who had experience with or interest in tracking period and fertility health data. We collected both quantitative and qualitative feedback and conducted a mixed-methods analysis. Our quantitative findings revealed that the usability of APFs contributed to reduced privacy risk, increased user control, enhanced trust, and improved usability of the apps’ overall privacy controls. From our qualitative findings, we identified several usability challenges in APFs, including a lack of transparency about these features, implicit approaches for seeking anonymity during onboarding, tedious and unintuitive process of skipping providing PII, and lack of feedback to evaluate the effectiveness of APFs. Based on our findings, we discuss the importance of improving the usability of APFs and propose actionable recommendations to address

identified challenges. These recommendations aim to enhance usability in PFT apps and provide insights for APFs in other contexts involving sensitive user data and data surveillance concerns.

Our contributions are: (1) We conducted a usability study to empirically understand how participants perceive and interact with APFs in PFT apps. (2) We identified prominent usability challenges in APFs in PFT apps. (3) We proposed actionable design recommendations to enhance APF usability of PFT apps and other sensitive-data contexts. (4) We laid an empirical foundation for future technical investigations by highlighting which APFs users notice, rely on, or misunderstand.

2. Related Work

2.1. Anonymity: Tools, Platform Features, and Usability Challenges

Anonymity, the ability to conceal one’s identity [1], [2], [3], is context-dependent and exists in degrees [1], [2], [3]. Specifically, from an attacker’s perspective, anonymity means being unable to sufficiently identify a subject within a group—known as the “anonymity set” [17]. Because both the attacker’s capabilities and the situational context can vary, anonymity must be considered within its specific social and technical contexts [17], [35].

A substantial body of research in S&P has examined anonymity through privacy-enhancing technologies (PETs), particularly tools like Tor. These tools aim to conceal users’ IP addresses and encrypt traffic to resist surveillance or tracking. Technical studies have focused on evaluating and improving the effectiveness of such tools under various threat models (e.g., [6], [7], [8], [9]). However, studies show that non-experts often misinterpret how Tor works [10], [36], have limited understanding of its operation and threat model [37].

Due to limitations of tools like Tor, everyday users often turn to built-in features provided by online platforms to manage their anonymity [4]. These include not disclosing real names or other identifiers in social networking, collaborative environments and other online contexts [4], [16], [38], [39]:

Anonymous or pseudonymous usernames. One common method is the use of anonymous or pseudonymous usernames, where individuals create usernames not tied to their real identities [13], [40]. For instance, 4chan allows users to post without accounts, and by default assign the name “Anonymous” when no username is provided [40]. Even pseudonyms can be re-used or claimed by other users in subsequent posts [40].

Anonymous posting or emailing. Some platforms support anonymous participation by allowing users to post content without registering an account or disclosing personal information [16], [41]. For example, Wikipedia enables what is referred to within its community as “anonymous” editing, allowing contributors to make edits without registering an account [41]. This lowers entry barriers for first-time

1. Femtech describes a sector focused on software, products, and services aimed at supporting women’s health [20], with an anticipated market value of \$50 billion by 2025 [21]. We recognize the term “Femtech” is problematic, as it overlooks transgender and nonbinary individuals [22].

contributors and encourages wider participation [16]. For anonymous emailing, services like Proton Mail does not require a phone number to create an account and allows users to adopt pseudonyms instead of their real names [14], [42]. Users appreciate that their activity is not easily linked to other email accounts and that IP logging is disabled by default [14].

Temporary or guest accounts. Some platforms offer temporary or guest accounts to facilitate access without requiring full registration [43], [44]. These accounts are particularly useful in sensitive contexts like mental health, where users value anonymity and reduced exposure [45].

These platforms’ built-in mechanisms support anonymity in ways that are often more accessible and contextually meaningful for everyday users. Importantly, they do not focus exclusively on technical identifiers such as IP addresses or data traffic [40], [41], [43], [44]. Instead, they often enable users to manage what personal information is shared—such as names, dates of birth, or email addresses. Prior work has noted that, for many users, anonymity is centered on concealing personally identifying details and managing how their identity is presented or withheld [4], [12]. This reflects a broader, user-centered understanding of anonymity that emphasizes control over identity disclosure in specific contexts.

Although prior research has explored why individuals seek anonymity through platforms’ built-in affordances, less is known about the usability of these features. Specifically, there is a lack of understanding regarding how built-in APFs affect users’ perceptions of privacy on these applications and how usable these APFs are. This gap is critical, as the usability of APFs directly affects user adoption and their ability to achieve intended privacy goals.

2.2. Threat Modeling in PFT Apps and the Rising Need for Anonymity

Privacy issues in PFT apps have drawn great attention in usable privacy research, especially after the overturn of *Roe v Wade* [24], [26], [28], [29], [32], [46], [47], [48], [49]. PFT apps often collect intimate health data, such as menstrual cycles and sexual activity, as well as behavioral data such as clickpaths and search histories [47], which can infer highly sensitive details like pregnancy status. Furthermore, these apps often collect identifying information like emails [48], [50], [51] which helps to link the sensitive health information with users’ identity. Despite the sensitivity of their user data, many PFT apps lack robust privacy protections [47], [48], [49]. For example, many PFT apps frequently share identifying information with third parties, like advertisers, data processors, and law enforcement [47], [48], [49]. Analyses of backend network transmissions reveal that such data sharing often spans multiple entities across different jurisdictions, with limited transparency about the scope and nature of these exchanges [31], [32]. This raises significant concerns about whether sensitive data is being handled appropriately, especially when transmissions cross legal boundaries [31]. Some PFT apps even transmit user

information insecurely, exposing users to interception and malicious exploitation [32]. Furthermore, researchers found that a PFT app claiming to offer an “anonymous mode” did not reduce IP address connections compared to default use, undermining its privacy claims [31].

Regulatory gaps further exacerbate these vulnerabilities. In the U.S., HIPAA protections apply only to clinical or medical settings, leaving PFT app data outside its scope [52], [53]. No comprehensive federal privacy law currently safeguards sensitive health data collected outside healthcare contexts [25]. In the UK and EU, ambiguity in GDPR enforcement regarding PFT apps creates potential risks, as users may assume their data is fully protected under “special category data” provisions, which are not always enforced [24], [54].

Threat modeling approaches applied to PFT privacy highlight diverse adversaries [27], [30], [55]. Corporate actors—such as app developers, advertising networks, and data brokers—actively seek to link reproductive data to user identities to monetize sensitive user profiles without meaningful user consent [25], [56]. For instance, embedding third-party trackers in apps and opaque data-sharing agreements facilitate identity linkage, enabling targeted advertising based explicitly on identifiable reproductive information, illustrating the monetization incentives driving these practices [27], [30]. The consequences of such practices extend beyond privacy violations. For example, receiving pregnancy-related targeted ads can cause significant emotional distress, particularly for individuals who have experienced pregnancy loss [25]. Users have also reported fears of social stigma, harassment, and emotional harm if sensitive reproductive data—such as irregularities in menstrual cycles—were linked to their identities and exploited by malicious actors [12], [29]. Employers and insurers might exploit identifiable reproductive health data through data-sharing channels to discriminate against individuals in employment, insurance coverage, or benefits decisions [25], [56]. Law enforcement also emerge as powerful adversaries, particularly in U.S. states where abortion is criminalized post-*Roe*, and where PFT data may be subpoenaed or purchased to identify and prosecute individuals [57], [58]. Moreover, legal scholars caution that the implications of the overturn of *Roe* extend beyond the U.S., potentially threatening reproductive health rights and privacy in other countries [59], [60].

Prior research has shown users’ strong desire for enhanced privacy protections within PFT apps [26], [27], [28], [29]. Specifically, there is a strong desire for robust privacy features, including enhanced data control, encryption, granularity, and anonymization [26]. In particular, many users emphasize the importance of anonymizing data before it is collected or shared [26]. Legal scholars echo these concerns, emphasizing the need to safeguard identifiable personal data in reproductive healthcare context. They warn that such data can end up with data brokers and be accessed by various parties, including advertisers and law enforcement [61], [62], underscoring the critical need for stronger anonymity and privacy protections. In response, some PFT apps emphasized their anonymous related features in public

communications—especially after the overturn of *Roe v. Wade*—as a means to maintain user trust [29], [47], [49]. While much research has examined the privacy practices of PFT apps [24], [32], [46], [48], few studies focus on the usability of their privacy features. One notable exception is Malki et al.’s work on the usability of privacy-related features in PFT apps, which used UI inspections to identify significant issues in account creation, privacy controls, and data deletion workflows [47]. They observed that registration processes often collected unnecessary data and failed to clarify when information was optional. The study also touched on APFs, such as an “anonymous mode” that stored only essential tracking data without linking it to identifiers like names, emails or technical identifiers. However, the usability of these APFs was not thoroughly investigated, leaving questions about their practicality.

A critical gap remains in understanding how users evaluate the usability of these features, particularly those related to anonymity. This gap is especially important in the post-*Roe* context [26], [27], [28], [29], where heightened privacy concerns and the need for anonymity play a crucial role in mitigating risks associated with PFT apps [47]. Addressing this gap is essential to inform the user-centric privacy designs that align with users’ needs and expectations for anonymity in the face of reproductive health surveillance.

3. Methods

To explore the usability of APFs in PFT apps, we conducted 32 usability testing sessions from March 29, 2024, to May 8, 2024, collecting and analyzing both quantitative data and qualitative data. This study received approval from our institutions’ IRB offices before data collection.

3.1. Participant Recruitment

To explore diverse privacy perceptions of PFT apps in the post-*Roe* era, we recruited participants from three states with differing legal landscapes [63]: Pennsylvania (PA), Florida (FL) and Texas (TX). These states were selected to ensure a broad range of perspectives. During the data collection period, PA permitted abortions up to 24 weeks of pregnancy [63], [64]. In FL, abortions were legal up to 15 weeks until May 1, 2024, when a six-week ban was enacted following a Florida Supreme Court decision [65]. In TX, abortion was banned in almost all circumstances, allowing private citizens to sue providers or individuals assisting those seeking abortions after approximately six weeks of pregnancy [63], [66].

Recruitment was done both online (e.g., local subreddits and Facebook groups) and offline (e.g., flyers in community spaces and school buildings), supplemented by personal networks and snowball sampling. Recruitment messages provided study details such as its goals, duration, compensation (Amazon gift cards at a rate of \$30 per hour), eligibility criteria (aged 18+, English-speaking, residing in the U.S., and having experience with or interest in tracking period and fertility data), and a link to a screener survey.

The survey helped select a diverse participant group based on demographics and tracking experiences, screening for race/ethnicity, gender, age, and PFT experiences.

We recruited 18 participants in PA, 10 in FL and 4 in TX. The stricter abortion bans in FL and TX may have deterred potential participants from discussing sensitive topics related to PFT apps. Privacy concerns and perceived legal risks in these states posed greater recruitment challenges compared to PA, where participants likely felt more comfortable sharing their experiences. The participant imbalance does not limit our study: our aim is to identify overarching themes that persist across different contexts, rather than to conduct state-by-state comparisons. While the post-*Roe* context provides important background, we focus on user perceptions of usability challenges of APFs in PFT apps and not state-specific legal implications. Participants were diverse in age, race/ethnicity, education, and PFT app experience. Most were aged 18–30 (16 participants) or 31–45 (15 participants), and all identified as female. Educational attainment was high: 17 held master’s, professional, or doctoral degrees; 7 held bachelor’s degrees. For details about their demographics and PFT experiences, see Appendix D.

3.2. Data Collection

We conducted 32 in-person usability testing studies. All sessions were conducted in English and audio-recorded, with app interactions being both screen- and video-recorded. Audio recordings were subsequently transcribed on Microsoft Word, then reviewed by the first author. Each usability testing session consisted of four parts:

(1) **Welcome & consent:** Participants were informed about the study and their right to withdraw anytime before completing a consent form.

(2) **Warm-up:** To explore participants’ understanding of personal information, we asked about what types of information they believe could identify them and how they conceptualize PII. Since anonymity involves the ability to conceal one’s identity [1], [2], [3], we used the concept of PII to help participants think about their potential identifiability, preparing them to engage with and discuss APFs in PFT apps. If participants were unfamiliar with the term “PII,” we would provide a working definition to them: PII refers to any data that can be used to identify an individual [67].

We kept the definition of PII broad and open-ended to encourage participants to share their own interpretations. This approach allowed for deeper insights into how participants conceptualize anonymity and identifiability in the context of PFT apps.

These warm-up questions established a baseline understanding of participants’ views on identifiability, helping us contextualize their behaviors during the tasks. When referencing “PII” in participant quotes and related analysis in the findings, we reflect their contextualized understanding of the term as discussed during the session.

(3) **App interaction tasks and follow-up surveys and interviews:** Participants completed tasks focusing on con-

cealing their PII on three systematically selected PFT apps, followed by post-task interviews. The systematic app selection process began with a broad search of the U.S. App Store and Google Play, yielding 53 apps that met baseline criteria (relevance, >1,000 reviews, free, English availability). From this pool, we selected apps that explicitly claimed and demonstrated support for user interactions without requiring users to disclose their identity. Specifically, we evaluated apps' use of anonymity-related terms (e.g., "anonymous" and "identify") and claims of APFs in their privacy communications (e.g., public statements and privacy policies), then assessed their features and interfaces to confirm if users could engage without providing PII. We used HIPAA's healthcare-specific PII definition, which includes identifiers such as names, email addresses, and credit card numbers [52], [68]. It is important to clarify that our focus was on apps that offer user-facing privacy features intended to help users seek, approach, or achieve anonymity, not assessing whether true anonymity was achieved on the backend (e.g., whether apps still collect potentially identifiable metadata), which is beyond this study's scope. This approach ensured a realistic focus on actionable APFs as experienced by users. The final review resulted in six apps: **Femometer**, **PD**, **Flo**, **Stardust**, **Femia**, and **Cycles**. Out of the six apps, each participant was assigned three apps they had not previously used to minimize familiarity bias. Details on the app selection process and assignments are provided in Appendices A and E. We designed two tasks to explore the APFs:

Task 1: You just downloaded this app and would like to track period and fertility-related data with it. Can you try to (1) set up the app and (2) track this data within the app, without providing any PII that could be used to identify you by the app?

Task 2: You have registered an account with PII to use this app to track period and fertility related data. Can you try to unlink the period and fertility data from your personal information without deleting the account completely, so your period and fertility data cannot be directly associated with your identity within the app?

These tasks offered a structured approach to assess APF usability. Task 1 focused on preventive measures to avoid PII disclosure during setup, while Task 2 addressed corrective actions to manage and unlink PII post-registration. Due to varying feature sets across apps, not all were compatible with both tasks. Only Flo supported post-registration unlinking of PII, and therefore was tested with both Task 1 and Task 2. The remaining apps were tested only with Task 1, as they did not offer functionality to decouple identity information after submission. Our study did not aim to isolate individual features through controlled experimental manipulations. Instead, we evaluated the usability of fully implemented APFs as they are presented to users in real-world contexts. As such, the variation in task applicability reflects actual differences in app capabilities. Each app served as a distinct design condition, where its specific combination of features and interface elements shaped how users understood and navigated anonymity. Capturing this

diversity was key to ensuring ecological validity in understanding user interactions with anonymity features.

We encouraged participants to think aloud to share their thought processes. During tasks, researchers observed non-intrusively, offering no hints or assistance. There were no time limits, and participants moved to the next step upon completing the task or indicating they could not continue. For detailed task designs, see Appendix E.

After completing the tasks, participants filled out a post-task survey on task success, app functionality, and ease of use. Follow-up interviews then probed their understanding of PII, anonymity concerns, challenges encountered, privacy strategies, and impressions of the apps' APFs.

Additionally, participants completed a post-app survey after engaging in task(s) within each app to measure app-level privacy attitudes and perceptions, and usability of the apps' privacy features. This survey included common privacy-related constructs frequently examined in privacy research [3], [69], [70], [71], [72], [73]: **Trust** (adapted from [74]), **Perceived Privacy Risk** (adapted from [74]), **Perceived Control over Privacy** (adapted from [73]), and **Usability of Privacy Settings/Controls** (adapted from [75]). Trust refers to the degree to which users feel that an app provider is dependable in protecting their personal information. Perceived Privacy Risk reflects users' expectations of potential loss associated with sharing their personal information with the app. Perceived control over privacy focused on participants' sense of agency over the disclosure and subsequent use of their personal information in the app, an important factor closely linked to privacy concerns. We also included the Usability of Privacy Settings/Controls construct, adapted from the System Usability Scale (SUS) [75] to evaluate how user-friendly participants found the overall privacy features in each app. We measured these constructs with a seven-point scale from strongly disagree to strongly agree. For details, please refer to Appendix B.

(4) Post-test interview: After completing interactions with the three assigned apps, participants took part in a post-test interview about their understanding of PII, experiences managing PII, and use of APFs across the apps. We also explored their broader experiences with PFT apps, including privacy concerns, behaviors, and strategies.

The study sessions ranged from 48 to 210 minutes. This variance reflects our exploratory study design: we did not impose strict time limits for task completion, allowing participants to proceed at their own pace, with some spending extra time exploring the apps when completing the tasks. For each app, we collected 15–17 individual data points.

3.3. Data Analysis

We conducted two types of analysis: a quantitative analysis on survey responses and a qualitative analysis of interview and think-aloud data during user tasks. For the quantitative analysis, we employed generalized linear mixed models (GLMM), cumulative link mixed models (CLMM), and multi-level structural equation modeling (SEM).

To answer RQ1, we employed GLMM and CLMM to compare the usability of APFs across apps, which are assessed with three DVs: **Success**: Perceived ability to track without providing PII — “Did you successfully track period and fertility-related data without providing PII?” (1 = Yes, 0 = No or Not Sure) **Affordance**: Perceived app support for anonymous tracking — “Do you think the app provides you with the function(s) to track this data without PII?” (1 = Yes, 0 = No or Not Sure) **Ease**: Perceived ease of tracking without PII — “Overall, how easy did you find it to track data without providing PII?” (7-point Likert scale, from “Very Difficult” to “Very Easy”). We fit the following models using R’s `lme4` and `ordinal` packages. For binary outcomes Success and Affordance, we built a GLMM:

$$\text{logit}(DV_{ij}) = \alpha + \sum_{k=2}^6 \beta_k \cdot \text{App}_k + u_i + \varepsilon_{ij} \quad (1)$$

For the ordinal outcome Ease, we used a CLMM:

$$\text{logit}(P(DV_{ij} \leq c)) = \theta_c - \left(\sum_{k=2}^6 \beta_k \cdot \text{App}_k + u_i \right) \quad (2)$$

Where:

- DV_{ij} is the response from participant i on app j .
- App_k are dummy-coded indicators for the tested six apps (Stardust is the reference category).
- β_k are fixed effects for each app relative to Stardust.
- $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is a random intercept for participant i .
- ε_{ij} is the residual error term (GLMM).
- θ_c are cumulative thresholds between ordinal categories (CLMM).

Post-hoc pairwise comparisons were conducted using the `emmeans` package with Holm correction. These models allowed us to assess pairwise differences between apps.

To answer RQ2, we constructed the multi-level SEM to examine how the usability of APFs influences users’ privacy perceptions in PFT apps. These perceptions include *trust* [74], *perceived privacy risk* [74], *perceived control over privacy* [73], and *usability of privacy settings/controls* [75].

Since trust, perceived privacy risk, and perceived control over privacy are latent factors, we first examined their convergent and discriminant validity in CFA using the robust weighted least-square estimator (WLSMV) to ensure the robustness of the measurement model. Convergent Validity was supported if all items loaded significantly on their corresponding factor with standardized factor loadings above 0.6, and if the Average Variance Extracted (AVE) was higher than 0.5. Discriminant Validity was confirmed if the correlation between factors was below 0.85 and less than the square root of the AVE for each factor. For usability of privacy settings/controls, we calculated its mean and standardized it for analysis. We chose not to include this variable in the factor analysis because usability, as measured by the SUS,

is typically treated via a sum score [76], [77]. Using the validated measurement model from CFA, we conducted multi-level SEM to evaluate how different usability metrics of APFs—Success, Affordance, and Ease—influenced privacy perceptions (risk, control, trust) and the overall usability of privacy controls.

To answer RQ3, we employed reflexive thematic analysis to analyze the qualitative data [78], [79]. Four authors reviewed recordings and transcripts to familiarize themselves with the data. They independently coded the data in Microsoft Word, examining quotes and assigning codes across the dataset. Concurrently, the team met to discuss initial codes, resolving disagreements and merging codes. Each data item received equal attention, and coding captured as many potential themes as possible [78], resulting in 274 initial codes. The team grouped similar codes to generate initial themes and iteratively refined them to ensure “internal homogeneity and external heterogeneity” [80]. Once finalized, themes were defined and named, integrating analysis with data extracts to create a coherent narrative of the findings. To preserve anonymity, participants were assigned pseudonyms (e.g., PA1, FL2, TX3), with the state abbreviation indicating recruitment site. Including the state in each pseudonym adds contextual insight into the participant’s legal environment while maintaining confidentiality.

3.4. Ethics Considerations

This study followed strict ethical protocols to protect participants’ safety, privacy, and well-being, particularly given the sensitive context of abortion and reproductive health in the U.S. All procedures were approved by the authors’ IRB offices, and written consent was obtained. Participants were reminded that they could withdraw from the study at any time without penalty and could skip any questions they preferred not to answer. During usability testing, researchers checked in regularly to ensure participants’ comfort when discussing PFT data and PII. Participants’ PII (e.g., email addresses) was collected solely for scheduling and compensation and was promptly deleted upon completion of compensation. To maintain confidentiality, all participant responses were anonymized and assigned pseudonyms during data analysis and reporting. Research materials are securely stored in password-protected databases, with permanent deletion planned after project completion. To prevent the disclosure of PII during the tasks, participants were provided with pre-created user profiles for use in the PFT apps to avoid sharing personal information.

3.5. Positionality Statement

Our team includes cisgender women and men with varying experiences in reproductive health privacy. The researchers conducting the usability testing sessions included the first three authors, all of whom identify as female. Researcher gender may have shaped participant experiences; having female facilitators may have helped participants (all self-identified as female) feel more comfortable discussing

sensitive topics related to reproductive health. The first, third, fourth, and last authors are long-term PFT app users, offering insider perspectives on functionality and privacy risks, while the second and fifth authors, with limited prior use of these apps, provide more external, critical views. As a team, we hold deep concerns about privacy in reproductive health, given the sensitivity of the data and the historical constraints on female bodily autonomy, which create a broad and urgent problem space for privacy protection [25]. Our research approach is guided by empathy for users, awareness of broader sociotechnical systems, and a commitment to reproductive health rights, informed by both our positionalities and experiences with PFT apps. In this study, we focus on the usability of APFs within PFT apps as part of the broader privacy ecosystem. While APFs shape how apps communicate privacy commitments and influence user expectations, their usability alone is insufficient for meaningful protection. Our focus on the APFs does not imply endorsement of the apps' design choices. For example, the mere presence of APFs does not justify sensitive data collection or making anonymity opt-in rather than the default.

4. Results

In this section, we present findings related to three RQs: (1) the quantitative results of usability of APFs between apps; (2) the quantitative results of how the usability of these features influences participants' perceptions of privacy and the overall usability of the app's privacy controls; and (3) the qualitative results of key usability challenges participants encountered when using APFs.

PFT apps typically allow users to log menstrual cycles, symptoms, sexual activity, and fertility-related data, while also offering predictions, reminders, and community forums. We found that during onboarding and use, these apps often encourage account creation, which involves sharing PII such as names or email addresses to store and synchronize user data. They also collect personal details such as birth year, height, and weight to support cycle analysis and prediction. To contextualize our findings, we first summarize how participants conceptualized anonymity in PFT apps. Their understandings varied: some (e.g., P8, PA16) perceived anonymity as storing data locally without linking it to an account, while others (e.g., PA3, PA15) emphasized the absence of PII collection and the inability to link in-app data back to the individual. Participants also identified specific data types they considered personally identifiable, such as email, phone number, name, birthdate, and address. Several also noted that individual attributes like height or weight may not be identifying on their own but could become so when combined with other descriptors (e.g., FL10, PA18). This finding aligns with prior work showing that users often understand how seemingly non-identifiable attributes can become identifying when aggregated [12].

4.1. Usability of APFs Between Apps

The usability of APFs in the apps were evaluated through (1) perceived successful tracking without PII, (2) app affordance for tracking without PII, and (3) ease of tracking without PII. As shown in Figure 1, participants reported the highest perceived success rates for tracking period and fertility data without providing PII on Cycles (93%), followed by Period Diary (PD; 81%) and Femometer (76%). In contrast, Stardust had the lowest perceived success rate (6%), indicating that users struggled most with its tracking features. The GLMM results show that participants were significantly more likely to report successful tracking on Femometer ($\beta = 6.03^{**}$), on PD ($\beta = 6.20^{**}$) Flo ($\beta = 4.58^{**}$), Femia ($\beta = 5.05^{**}$) and on Cycles ($\beta = 8.75^{**}$) compared to Stardust.

In terms of perceived app affordance for tracking without PII, participants reported the highest perceived affordance on Femometer and Cycles, with 94% and 93% of participants, respectively, agreeing that the app provided the function for tracking without PII. PD followed with 81%, while Stardust had the lowest perceived app affordance (18%), suggesting limited support for privacy-preserving tracking features. GLMM results showed that participants were significantly more likely to perceive Femometer ($\beta = 27.31^{***}$), PD ($\beta = 16.28^{***}$), Flo ($\beta = 15.59^{***}$), Femia ($\beta = 17.91^{***}$), and Cycles ($\beta = 27.62^{***}$) as affording tracking without PII compared to Stardust.

As for ease of tracking without PII, Cycles emerged as the leader in ease of use of APFs (Mean = 6.43), followed by Femometer (Mean = 6.18). Stardust scored the lowest again (Mean = 2.38), highlighting significant usability challenges with its APFs. CLMM results showed that participants rated the ease of tracking without PII significantly higher for Femometer ($\beta = 5.06^{***}$), PD ($\beta = 4.97^{***}$), Flo ($\beta = 3.12^{***}$), Femia ($\beta = 3.91^{***}$), and Cycles ($\beta = 6.30^{***}$) compared to Stardust. Additionally, post-hoc pairwise comparisons show that Cycles was rated significantly easier to use than Flo ($Estimate = 3.17^{**}$).

Summary: Cycles outperformed all other apps across all usability metrics for APFs. Femometer and Flo also performed well, ranking below Cycles but above the remaining apps. Stardust consistently received the lowest scores.

4.2. How Usability of APFs Impact Privacy Perceptions of PFT

Since privacy perceptions of PFT, namely perceived privacy risk, perceived control over privacy, and trust, are latent factors measured by multi-item scales, we first reported their validity before examining the effect of usability of APFs on them. The CFA results show that our measurement model's convergent and discriminant validity were supported. First, all factor loadings exceeded the threshold of 0.6 [81] (see "Factor Loading" column in Appendix B). Each latent variable's AVE was also above 0.5, indicating strong convergent validity. In addition, discriminant validity was supported, as correlations between factors were all below 0.85 and also

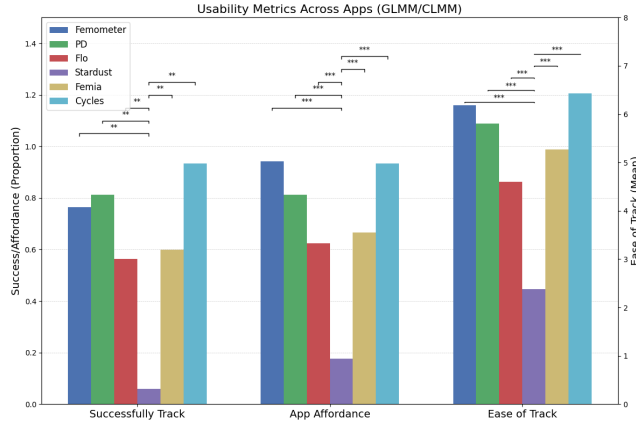


Figure 1. Comparison of usability metrics for APFs across apps. Bars show the proportion of participants who responded “Yes” for *Successfully Track* and *App Affordance* (left y-axis, ≤ 1), and the mean Likert rating for *Ease of Tracking* (right y-axis, > 1). Asterisks above lines indicate significant pairwise differences between apps based on GLMM and CLMM post-hoc comparisons with Holm correction. Statistical significance levels: *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

smaller than the square root of each AVE, as shown in Appendix C.

Figure 2 illustrates that the usability of APFs significantly influences users’ perceived privacy risk, perceived control over privacy, trust, and the overall usability of the app’s privacy features and controls. According to the SEM results, participants who successfully tracked period and fertility data without PII (the red bar in the ‘Risk’ category of the top panel in Figure 2, $\beta = -0.6^*$), perceived the app as providing affordances for tracking data without PII (the red bar in the ‘Risk’ category of the mid panel in Figure 2, $\beta = -0.6^*$), and found the tracking without PII features easy to use ($\beta = -0.6^{**}$) reported significantly lower perceived privacy risk compared to the baseline groups (blue bars in ‘Risk’ category of Figure 2).

Participants who successfully tracked data without PII ($\beta = 0.6^*$), perceived the app as providing affordance for tracking data without PII ($\beta = 0.6^*$), and found the tracking without PII features easy to use ($\beta = 0.9^{**}$) or of medium ease to use ($\beta = 0.6^*$) reported significantly higher perceived control over privacy compared to the baseline groups.

Participants who successfully tracked data without PII ($\beta = 0.5^*$), perceived the app as providing affordance for tracking data without PII ($\beta = 0.5^*$), and found the tracking without PII features easy to use ($\beta = 0.7^{**}$) reported significantly higher trust compared to the baseline groups.

Participants who successfully tracked data without PII ($\beta = 1.0^{***}$), perceived the app as providing affordance for tracking data without PII ($\beta = 0.7^{**}$), and found the tracking without PII features easy to use ($\beta = 1.1^{***}$) reported significantly higher perceived usability compared to the baseline groups.

Summary: Usability of APFs in the apps contribute to reduced privacy risk, increased control, enhanced trust, and improved usability of the apps’ overall privacy controls.

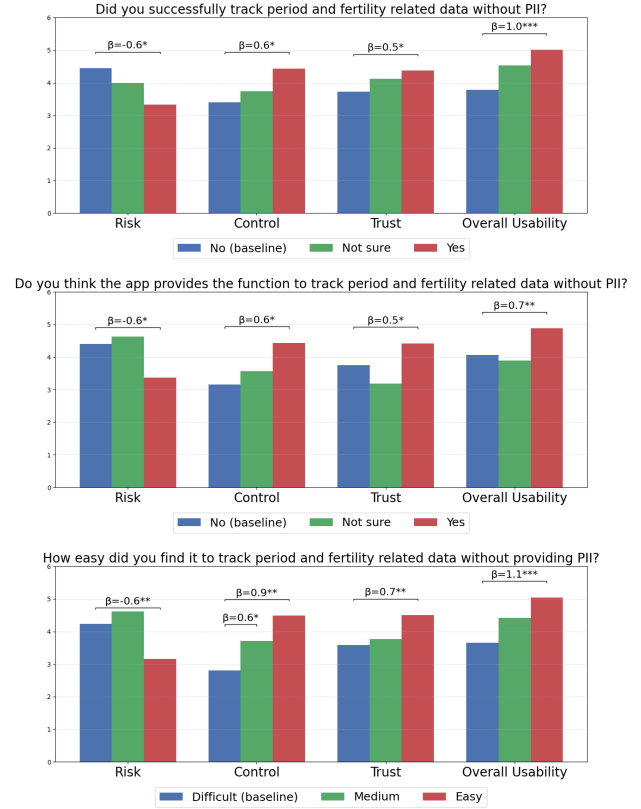


Figure 2. Usability of APFs and their impact on users’ privacy perceptions and the usability of privacy controls. Connecting lines indicate statistically significant differences compared to the baseline group (blue bars). The β coefficients are derived from regressions with dependent variables: Risk (perceived privacy risk), Control (perceived control over privacy), Trust, and Overall Usability (usability of privacy settings/controls). Statistical significance levels: *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

4.3. Usability Challenges in APFs

In this section, we present the prominent identified usability challenges in APFs, including unclear communication about APFs, implicit approaches for anonymity during onboarding, tedious process of skipping providing PII, and lack of support to evaluate the effectiveness of APFs².

4.3.1. Lack of Transparency about APFs. The lack of clear communication and poorly integrated privacy features emerged as a prevalent usability challenge, leaving users uncertain about their data protection. Below, we examine examples from Stardust and Flo to highlight how these issues arise and the design choices driving user confusion and mistrust.

Poorly Communicated Anonymity by Default. As shown in 4.1, Stardust scored the lowest in perceived successful tracking, perceived app affordance, and perceived

2. Because data collection occurred in Spring 2024 and app interfaces might evolve over time, we revisited the apps in August 2025. While some interface elements had changed, our findings on APFs remain valid—for example, the screenshots in 4.3 are still representative.

ease of use of APFs, indicating significant usability challenges with its APFs. The qualitative data suggests that these challenges are primarily due to a lack of transparency and unclear communication about these designs. While Stardust’s privacy policy (as of May 2024 during study phase³) outlines its approach to anonymity, stating that user PII (e.g., email and phone numbers) is managed by a third-party and not directly linked to period data, this is not communicated effectively within the app. In their privacy policy, Stardust emphasizes that in cases of subpoenas, it cannot connect period data to user login information. However, this critical detail is buried within the privacy policy’s legal text, subtly placed in small font at the bottom of the page, as shown in Figure 3(a). As a result, most participants (16/17) remained unaware of these anonymity affordances.

Only one participant (PA2) who reviewed the privacy policy expressed confidence in their understanding of Stardust’s APFs: “[Stardust makes you feel you are anonymous because they don’t have your login data]. They told me in the policy that they don’t [have] my name with my data. They do have my [period] data, but even if the government asked them, they can’t tell them this [period] data is for this person, so I think [...] I am anonymous always in this [app].” PA2 believed she was anonymous despite providing PII like her email address because the app cannot link her health data back to her identity. However, this understanding was an outlier among participants. The majority struggled to reconcile the app’s claims of anonymity with their onboarding experience, which required the input of PII. As shown in Figure 3(a), Stardust’s welcome screen offers no indication of its APFs. Instead, users are immediately prompted to provide their email or sign in using Apple credentials. This creates a sense of lost control over their data. For example, PA15 described her experience: “[I felt I lost control over my data] when I was trying to explore [Stardust] and then I had the key in. I [have to provide an] e-mail.” The perception of lost control during onboarding, coupled with the absence of any explicit mention of APFs, highlights a missed opportunity for the app to educate users about its privacy affordances. This lack of transparency likely contributes to users’ difficulties in perceiving the app’s anonymity capabilities, which might cause privacy concerns in sensitive post-Roe contexts.

“Hidden” Anonymous Mode. Flo offers an anonymous mode that, according to its privacy policy, allows users to “access the app without your name, email address, or technical identifiers being associated with the data you put into the app.” While commendable for enhancing user privacy, the feature is buried in the account settings and is poorly mentioned during onboarding or other key interactions, leaving many participants unaware and highlighting a critical usability issue. As shown in Figure 3(b), users must click on a banner-like element (the rectangular section displaying the profile avatar and email address) in the settings page to access anonymous mode. However, the banner does not

visually indicate that it is clickable, leading many participants to overlook it. One participant, PA6, explained: “*This [banner for opting in anonymous mode] doesn’t seem like a place that you could click [...] so I was never gonna go into here to get to where I need.*” PA6’s feedback highlights that the interaction is unintuitive, making it difficult for users to manage their anonymity-related privacy settings. This lack of clarity undermines the usability of a feature that is critical for privacy-conscious users.

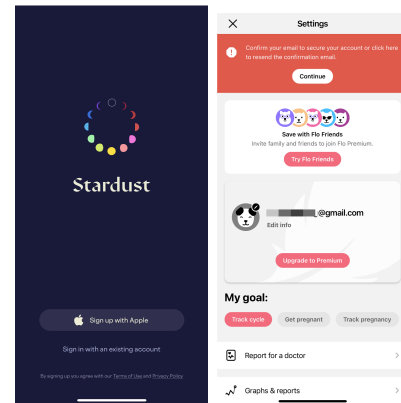


Figure 3. Screenshots from two apps: (a) Stardust’s welcome screen and (b) Flo’s settings page.

Such poor usability design may even lead users to question the app’s privacy intentions, ultimately damaging their trust. Some participants speculated that the anonymous mode’s lack of visibility might be intentional, suggesting that the app’s design encourages users to remain linked to their accounts. For example, participant FL2 remarked: “[I think a] Pencil button [that you can click to opt-in to anonymous mode] could be more [obvious]. But I think they will never do that because [the app] [wants] you to link the account so that they can get more information [about you].” This perception suggests that the lack of transparency may cause users to doubt the app’s genuine commitment to privacy and can even be experienced as privacy-related deceptive designs (also known as dark patterns)—design choices that increase friction for privacy-preserving actions while nudging users toward data-sharing defaults [82], [83]. Users may interpret such poor usability of APFs as intentional obfuscation, further undermining trust in the app’s privacy practices.

Summary: Poor transparency, unclear communication, and buried privacy settings regarding anonymity lead to user confusion and undermine trust in the apps’ privacy commitments.

4.3.2. Implicit Approaches for APFs During Onboarding. This section examines how implicit versus explicit approaches to anonymity during onboarding impacts users’ understanding and trust in APFs. We compared Flo’s implicit design—where users must infer available options—with PD’s explicit interface that visibly offers anonymity as a

3. <https://web.archive.org/web/20240525165937/https://stardust.app/privacy-policy.html>

selectable feature. This comparison highlights the usability challenges associated with implicit APFs.

Implicit Skipping as a Path to Anonymity. Participants noted that Flo adopts an implicit approach to APFs during onboarding, making them harder to identify and understand. As shown in Figure 4(a), Flo’s onboarding process does not prominently present anonymity as an active choice. Instead, users are required to skip providing account information by clicking a small “X” at the top of the screen. This design relies on users recognizing the skip function and correctly inferring its implications on anonymity and privacy, which can lead to confusion. Although Flo does provide an anonymous mode embedded in privacy settings that users can opt-in to, as described in Section 4.3.1, the onboarding process—which often shapes participants’ first impression of the app’s approach to privacy—is crucial for communicating APFs clearly.

Flo’s implicit approach during onboarding leaves participants feeling as though they must navigate around providing their personal information, which may inadvertently compromise their anonymity. For example: *“Flo almost seems like they wanted you to create the account and have that information in there. It gave you the option to make it anonymous, but it was not nearly as obvious that [it] was available [for] you to do [compared to] PD [where you can actively opt-in to being anonymous]”* (PA6). The participant underscores that Flo’s approach to avoiding account creation and providing PII during onboarding is unclear and implicit, making anonymity feel less accessible, as she must infer that skipping is the way to maintain her privacy. This lack of explicit guidance creates confusion and requires users to take indirect actions to protect their privacy. She also expressed a strong preference for a more proactive and obvious approach to promoting anonymity during onboarding.

Preference for Explicit Opt-In for Anonymity. In contrast, PD uses an explicit approach to anonymity by clearly offering a “Use anonymously” option during onboarding. Unlike Flo’s approaches, where users must navigate around providing PII or infer how to avoid it, PD takes a proactive approach by prominently offering “Use anonymously” as a parallel and visible option alongside account creation methods such as Google, Facebook, or email, early in the onboarding process (Figure 4(a)). PA7 expressed a strong preference for this approach: *“PD was just very explicit of [the anonymous mode]. Typically, I would be looking for how can I log in... But when I saw the ‘use anonymously’, I knew that was the right one to click. [...] And I think for this one it came up pretty close to the beginning... before I had to put too much [information] in [the app].”* PA7 highlights the strength of PD’s explicit design approach, which contrasts with more implicit designs like Flo’s. This approach is particularly effective because it directly aligns with user expectations: users are not required to infer their options or navigate away from the default flow. Instead, the anonymous mode is clearly stated and positioned, making it easy and intuitive to select.

Summary: Implicit onboarding flows that require users to infer or skip steps in order to maintain anonymity obscure

the availability and function of APFs, reducing their effectiveness and increasing the risk of unwanted disclosure. In contrast, explicit designs that clearly present these options enhance usability, trust, and privacy.

4.3.3. Tedious and Unintuitive Process of Skipping Providing PII. In this section, we illustrate how design flaws like cumbersome skipping processes and poorly placed skip buttons create usability challenges for users seeking anonymity.

Burdens of Skipping Providing PII: Participants frequently highlighted that some apps, such as Flo and Femia, ask numerous questions during app setup, some of which may potentially involve PII. While users have the option to skip many of these questions, the process of skipping was described as tedious, unintuitive, and time-consuming. This friction in the setup process undermines the very goal of enhancing user anonymity. For example, PA10 shared: *“[Flo’s] Do it without PII part [is] just tedious... [Skipping] still took time. It would just be longer to do it when you’re providing your information [than skipping].”* PA10 expressed frustration with both providing PII and skipping it, describing both options as burdensome. While skipping was slightly faster, it still required significant effort. The setup process offered options for skipping PII and preserving anonymity, but it ultimately took more time than participants expected, reducing overall usability and satisfaction.

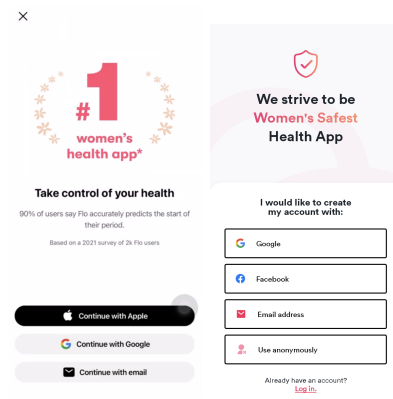


Figure 4. Account creation pages from (a) Flo and (b) Period Diary (PD).

In addition, participants highlighted challenges with the usability of the skip feature. For example, PA9 noted that the skip button often applied to only a single question rather than allowing users to skip the entire process: *“I look for an X... instead of the Skip button [to] skip [it] all together. So it’s kind of like a prefer to answer [rather than skip the question].”* PA9 noted that the limited functionality and the overall burden of skipping multiple questions involving potential PII may discourage her from trying to skip providing PII. This suggested that if anonymity feels too inconvenient or impractical, users might forgo it entirely.

Participants suggested streamlining the setup process by reducing the number of questions or offering greater user choice in how to proceed anonymously. For example, P11

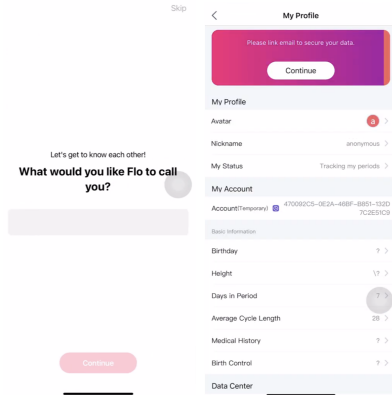


Figure 5. Screenshots from (a) Flo's setup process and (b) Femometer's data overview page.

noted “I feel like they could probably dumb down [some] questions, [Currently] it's a lot. I feel like if you wanted to enhance more on what you're getting from the app, you could probably give a section for it, [where] the person can make a choice and be like ohh I'll answer some more questions to make the app more valuable [and] beneficial for me to use.” This suggestion reflects a preference for a tiered approach to app setup. Users could initially face only essential questions, with the option to answer additional questions later if they wish to enhance app personalization. Such an approach could strike a balance between personalization and privacy, offering users a smoother and less frustrating experience to maintain anonymity.

Skip Button Usability. The design and placement of the skip button significantly influence how users perceive their ability to maintain anonymity during the setup process. For example, a participant shared: “So some of the questions, there were like 6 different options. I think the skip button was up here, so I just didn't even think about [looking for the skip button there]. I didn't notice it until the very last questions. And I'm like, oh, I wonder if it was there the whole time.” (PA9) As shown in Figure 5(a), the skip button was not immediately noticeable due to its positioning or design, preventing users from recognizing it as an option.

Participants highlighted that the design made the skip option feel less accessible, which gave the impression that they had fewer choices or were obligated to provide answers. For example, PA6 noted: “The skip was up in the corner. It's not quite as obvious that you have that option. You might feel like you have to choose some of these options to move forward.” This lack of visibility for the skip option affects the user's perception of autonomy and control over their data. When users feel forced to answer questions due to unclear or hidden alternatives, it undermines trust and might even frustrate their attempts to remain anonymous.

The unclear presentation of the skip button also led some participants to question the app's motives, particularly when apps requested personal details like a name. For example, TX4 expressed skepticism when encountering Femia's request for a preferred name despite the presence of a skip

button in the corner, remarked: “They really want to identify [me].” This illustrates how poor APF usability can heighten concerns that apps prioritize data collection over privacy. The hidden or unclear presentation of the skip button reflects patterns documented in prior work on deceptive design patterns like *obstruction* and *interface interference*, where privacy-preserving choices are made less salient or harder to access than data-sharing defaults [84]. Echoing findings on cookie consent popups [82], our participants felt they had “no real choice” but to provide data, despite the technical availability of a skip option. This suggests unusable APFs can act as privacy-related deceptive designs, undermining autonomy and trust.

Summary: Tedious and unintuitive PII skipping processes, coupled with poorly designed skip buttons, undermine usability of APFs and even erode user trust.

4.3.4. Lack of Feedback to Evaluate the Effectiveness of APFs. This section highlights a key usability challenge: the lack of feedback that enables users to evaluate the effectiveness of APFs, which burdens users in assessing their anonymity and hinders their efforts to maintain it.

Lack of Visibility into Collected Personal Data. While participants valued APFs that allow them to withhold their information from apps, participants also emphasized the importance of evaluating whether these features are effective. Specifically, users expressed a need for feedback on what information had been collected and greater transparency about their anonymity status. A lack of such feedback often left participants feeling disappointed and distrustful of the app. For example, PA9 noted: “Cycles just doesn't [provide the overview of what has been collected from me]. I don't know where to find any of my information... it just doesn't give me as much trust because I don't even know where to find my information.” This participant's frustration stems from the inability to verify what information is being collected or stored, which makes it difficult for them to assess their anonymity within the app. Although Cycles collects minimal information during onboarding (e.g., no name or email), which contributes to its favorable usability score for APFs (as noted in 4.1), the lack of clear and visible feedback on collected personal data still raises concerns for users who may fear that the app possesses information they are unaware of.

Preference for Clear Feedback on Anonymity Status. Another app, Femometer ranked among the top three apps in all metrics of usability for APFs, achieving the highest proportion of participants who responded “Yes” for app affordance and highest mean score for ease of use (Section 4.1). The qualitative data sheds light on this strong performance, highlighting that the app provides users with a clear and comprehensive overview of the data it has collected. This transparency reassures users, as it demonstrates that the app is not collecting or hiding unnecessary information. For example, PA9 praised Femometer for its clear data display: “Femometer is more straightforward [in terms of displaying data collection], it just shows you straight up like it's anonymous and then you have no information like it sort of shows

you.” As is shown in Figure 5(b), this explicit presentation of user data allows users to verify their anonymity status at a glance. By showing both the information that has been collected and confirming the absence of additional personal information, Femometer reduces concerns about hidden data collection.

Summary: A lack of feedback on collected personal data and anonymity status hinders users’ ability to evaluate APFs, leading to frustration and distrust.

5. Discussion

5.1. Usability Challenges of APFs in PFT Apps

Fragmented User Journeys for Anonymity. Participants reported that anonymity-related decisions were fragmented across multiple touchpoints in their interaction with PFT apps. Unlike some general privacy settings, which often involve straightforward, one-time choices (e.g., enabling or disabling location access or agreeing to a privacy policy), APFs require more active engagement. For example, our participants make anonymity-related decisions throughout their entire engagement with the app. These decisions include whether to create an account during onboarding, whether to share or withhold certain pieces of information when the app requests specific data, and how to unlink or pseudonymize their data during ongoing app usage. This fragmented design requires users to navigate a complex and discontinuous process, demanding more effort and awareness compared to cohesive privacy controls.

Such fragmentation undermines effectiveness. Privacy is often secondary to users’ primary goals [85], [86], such as tracking period data in PFT apps. Furthermore, users have limited cognitive resources to evaluate all options and consequences of their actions [85]. They expect privacy protections to function seamlessly in the background, requiring minimal effort or attention. APFs in these apps often demand continuous attention and repeated decision-making, forcing users to re-evaluate anonymity choices across scattered controls and increasing cognitive load. This burden is compounded by the complexity of privacy tradeoffs, such as weighing the benefits of sharing data for functionality against the risks of losing anonymity. Fragmented or unintuitive APFs make configuration difficult, often leading to frustration or abandonment, even among users who prioritize anonymity. As [86] warned, “If security is too difficult or annoying, users may give up on it altogether.” For instance, PA9 described the tediousness of repeatedly skipping identity-related questions, which discouraged her from withholding PII.

Lack of Transparency about Anonymity Mechanisms. Our findings demonstrate that even technically robust anonymity mechanisms can fail to instill trust if they are poorly communicated, difficult to access, or unintuitive to use. For example, Stardust, despite providing anonymity protections by decoupling personal identifiers from sensitive data, scored lowest across usability metrics because participants were unclear how its settings supported anonymity

(e.g., PA15’s confusion). Even when apps attempt to communicate their APFs, failing to do so in a clear and accessible manner or to integrate these features seamlessly into the user experience undermine their effectiveness. For example, Flo emphasized its commitment to anonymity in privacy communications with users [49], [87], yet participants like FL2 found its anonymity mode hard to locate and suspected intentional obfuscation. Such design failures create friction, erode trust, and can resemble deceptive patterns that nudge users toward data-sharing defaults [82], [83], [84].

Inconsistent APFs Across Apps. We found that PFT apps adopt significantly different approaches to anonymity, reflecting a lack of industry standards in APFs, as highlighted by usability heuristics on consistency and standards [88]. For instance, Cycles avoids requesting PII entirely, while Flo offers an opt-in anonymous mode that helps users unlink their health data from identifiers. This diversity might cater to different user needs (e.g., some may favor default-anonymous experiences, while others may value explicit opt-in as deliberate control), which can be commendable given that users often have varying privacy preferences [89]. However, this variability can leave users uncertain about what to expect or how to navigate APFs, especially when apps fail to clearly explain how their anonymity mechanisms work. As supported by our screening survey results, users frequently try multiple apps before settling on one or switch over time. Transitioning between apps with fundamentally different APFs can be confusing, especially when users are accustomed to a specific approach. Without clear communication, this variability imposes cognitive burdens, creating confusion and frustration, ultimately undermining users’ ability to achieve or trust anonymity in a fragmented design ecosystem.

Critical Need for Usability of APFs in Reproductive Health Surveillance and Other Sensitive Contexts. APF usability is especially critical in sensitive contexts like reproductive health surveillance, where the stakes are high and the risks are deeply personal. In the post-*Roe* era, concerns about PFT apps have escalated, with fears that sensitive data could be misused in abortion-related prosecutions or subpoenas [57], [58]. Users also risk emotional harm, stigma, or harassment if cycle irregularities are exposed to malicious actors or intimate adversaries [29]. Threat modeling underscores multiple adversaries—law enforcement, advertisers, and anti-abortion groups—exploiting vectors such as third-party tracking, opaque data sharing, and weak access controls [27], [30]. Moreover, the implications of *Roe*’s overturn extend globally, threatening reproductive privacy beyond the U.S. [59], [60]. This global dimension further underscores the urgency of safeguarding intimate health data and highlights the importance of usable privacy designs in PFT apps. While prior research has emphasized users’ need for APFs in PFT apps [26], our study emphasizes the critical role usability plays in improving effectiveness and users’ privacy perceptions of the apps. Anonymity protections that are poorly designed, confusing, or hidden from users fail to support meaningful threat mitigation. Furthermore, our findings on hidden or obscured APF options align with research

on dark patterns and deceptive designs that nudge users toward data-sharing defaults [82], [83], [84]. In sensitive domains such as reproductive health, these tactics can disproportionately harm vulnerable populations by exploiting trust and cognitive load, leading to unintended disclosure [90]. Collectively, this underscores the need for APFs that are not only technically sound but also transparent, accessible, and respectful of user autonomy.

Moreover, the implication of our findings extends beyond reproductive health to other sensitive domains, such as non-traditional dating platforms in contexts where such identities may be criminalized. In such contexts, inadequate anonymity protections can likewise compromise user rights and expose individuals to third-party surveillance [91], [92]. For example, our findings on APF usability challenges echo prior work showing how closeted LGBTQ+ dating service users must adopt deliberate strategies—such as withholding personal details or obscuring visual cues—to preserve anonymity in the absence of sufficient and usable platform safeguards [91], [92]. This underscores that well-designed, usable APFs are essential in high-stakes contexts where privacy harms are severe and platforms bear a heightened responsibility to safeguard user identity.

Enhancing Privacy-by-Design and Privacy-by-Default in APFs. Many PFT apps share user data with third parties with little transparency and, in some cases, transmit information insecurely, exposing users to serious risks [31], [32], [47], [48]. These practices illustrate a broader failure to adhere to privacy-by-design principles, as emphasized in GDPR [93]. Our findings further suggest that front-end usability problems compound these technical shortcomings: even when APFs are present, poor design leaves users uncertain, frustrated, and ultimately vulnerable. Relatedly, privacy-by-default requires that personal data be processed with the highest level of protection [93]. By default, only data strictly necessary for a given purpose should be gathered, with privacy-protective settings enabled automatically rather than left for users to configure. From this perspective, simply offering APFs does not justify the collection or retention of sensitive reproductive health data or PII, and requiring users to opt in to anonymity is inadequate in today’s heightened reproductive privacy landscape. Developers’ responsibilities therefore extend beyond providing APFs to ensuring safe data practices consistent with GDPR principles such as data minimization and privacy by default. Furthermore, because our lab-based usability testing emphasized usability flows, participants’ feedback in our findings centered on interface experiences rather than the full spectrum of privacy concerns. For instance, PA7 recognizing the “use anonymously” button as “the right one to click” did not imply confidence that the feature truly safeguarded privacy. Consistent with prior work on user privacy concerns [12], [26], [29], we stress the need for a comprehensive approach that couples robust backend safeguards with usable front-end features, ensuring privacy-protective defaults are informed, accessible, and effective.

Importantly, while PFT app companies have clear in-

centives to improve APFs to retain users post-*Roe*, they also face strong disincentives: many apps rely on data flows that conflict with privacy-by-design and privacy-by-default principles [25], [27], [30]. For example, embedding third-party trackers can facilitate identity linkage and targeted advertising, reinforcing monetization incentives [27], [30]. When revenue models depend on expansive data collection and sharing, companies may keep APFs less visible, less usable, or opt-in rather than default. Addressing this tension requires more than app-led feature improvements. Independent privacy audits, systematic reviews of third-party SDKs, and vendor contracts restricting secondary use are essential to reducing linkability risks. Complementary measures—such as stronger regulatory enforcement against deceptive consent flows, app-store policy pressure, media scrutiny, and greater public awareness—can realign incentives by raising the cost of opaque tracking and rewarding privacy-preserving models.

5.2. Design Implications for APFs

This section presents recommendations for improving APFs, particularly in PFT apps, where users share sensitive personal information in surveillance-prone environments. Although grounded in the PFT context, our findings extend more broadly: the design implications we identify apply to APFs in other digital systems that handle highly sensitive data and are not specific to PFT apps.

5.2.1. Designing Effective Onboarding for Anonymity.

The onboarding process is a crucial touchpoint for introducing and promoting APFs.

Introduce APFs Early. Apps should provide a clear guide of their APF during onboarding as participants often felt confused when these options were not introduced early. For example, PA15 assumed Stardust lacked APFs because it immediately requested an email without explanation, undermining trust and a sense of control. To address this, apps should include a dedicated onboarding screen that explains available APFs and their mechanisms in accessible language. This proactive approach builds trust and ensures users understand their options. Aligning with design principles of visibility and transparency [94], such messaging enables users to make informed decisions about anonymity from the start.

Make Anonymity a Visible Choice. Anonymity should be presented as a clear and explicit option during onboarding. Participants, like PA7, praised PD’s proactive approach, where anonymity was visibly offered as an option (e.g., “Use Anonymously”) alongside traditional login methods. In contrast, Flo’s implicit approach required users to infer their options, such as skipping account creation, leading to confusion and making anonymity feel like an afterthought. To address this, apps should actively present anonymity options using clear, action-oriented language (e.g., “Enable Anonymous Mode”) to reduce cognitive load and minimize confusion, ensuring users can easily make informed choices.

Streamline Skipping PII Input. Skipping PII should be streamlined to reduce frustration and improve accessibility. Participants found the process tedious, requiring repeated actions that may increase errors and discourage anonymity-seeking. In PFT apps, fertility-related inputs such as birth year were perceived as both identity-revealing and burdensome to skip, even when justified as necessary for cycle analysis. This underscores the need for clear communication about why data are collected and whether they are essential or optional, so users can make informed choices without stress. Apps can improve usability by offering options like “Skip All” to bypass non-essential PII in one step or “Group for Later” to defer inputs to an optional section. Clear confirmations and consistently placed skip buttons aligned with standards can further minimize cognitive load, improve discoverability, and encourage engagement with APFs.

5.2.2. Anonymity Needs a Comprehensive and Cohesive Design Approach. Anonymity is not binary nor achievable through a single feature. As Nissenbaum [2] notes, it exists in degrees and through layers of cloaking. Our findings support this view: participants made anonymity-related decisions throughout onboarding and everyday use, underscoring anonymity as a continuous, context-dependent process rather than a one-time configuration. This ongoing nature of anonymity requires a comprehensive and cohesive design approach, where features—such as optional disclosure, identity visibility controls, pseudonyms, and account unlinking—are intentionally integrated and consistently reinforced. When aligned across the user journey, these features better reflect how users experience anonymity: flexibly, incrementally, and across changing contexts.

While prior S&P work has focused on protecting technical identifiers (e.g., IP addresses, network metadata), our study shows that users often equate anonymity with controlling the visibility of a broader range of PII. This includes not only technical identifiers but also commonly recognized attributes like names, birthdates, and email addresses. For many, anonymity is less about technical traceability than about managing identity boundaries—deciding when, where, and how they appear to others or the platform itself. To support this more holistic and user-centered conceptualization of anonymity, platform features—such as anonymous or pseudonymous usernames, the ability to skip providing PII, unlinking accounts, or toggling identity visibility—must function as a cohesive system. When integrated across the app lifecycle, these features reinforce one another, reducing confusion, improving usability, and giving users more intuitive, flexible control over their anonymity.

Clear Communication About Anonymity Levels. Apps must clearly define and communicate the level of anonymity they provide to avoid misconceptions or mistrust. Anonymity is context-dependent, not an “all-or-nothing” quality [2], [17]. The degree of anonymity a user experiences depends on what types of data are collected, retained, or linked to their identity, making transparent disclosure essential—ideally in accessible, non-technical language that explains key data practices. For example, while some apps

stated that it does not collect PII like email or name, they did not clarify how device IDs are handled, creating obstacles for users to evaluate their anonymity. In contexts involving especially sensitive data—such as PFT apps—clear, contextualized communication is even more critical. Users navigating reproductive health surveillance risks need explicit information about what “anonymity” means in that particular app, including potential protections (or limitations) against adversaries such as governments, data brokers, or third-party trackers. More broadly, platforms in domains like mental health, political expression, or support for marginalized communities would also benefit from offering transparency around how anonymity is structured and where its boundaries lie.

Centralized Anonymity Management. Apps should offer a centralized “Anonymity Controls” hub to streamline settings such as unlinking data, enabling anonymous mode, and reviewing status. Proactive alerts could flag actions that compromise anonymity (e.g., linking an email) and suggest mitigations, simplifying usability and boosting user confidence. Proactive alerts could flag actions that compromise anonymity (e.g., linking an email) and suggest mitigations, simplifying usability and boosting user confidence. This is especially valuable in PFT apps, where sensitive health data require frequent review as privacy risks evolve. More broadly, centralized controls would benefit platforms like anonymous email services or pseudonymous online communities, where users must dynamically manage PII visibility in response to shifting threats. A well-organized hub reduces cognitive burden, improves usability, and empowers users to control identity exposure.

Proactive Alerts for Anonymity-Related Risk Mitigation. Real-time alerts can help users act immediately to protect anonymity. For example, if a user links an email to an anonymous account, the app could warn: “*Linking your email may reduce anonymity. Unlink now?*” Such alerts should not only inform users of risks but also provide simple, actionable options. As timing is a critical dimension of effective nudging [85], just-in-time alerts, delivered when a privacy-compromising action is about to occur, reduce reliance on user inference and facilitate decision-making [85]. Additionally, context-specific notifications can highlight vulnerabilities and guide proactive adjustments. For example, an app might warn: “*Your location data may be subpoenaed. Disable tracking to enhance anonymity?*” Such tailored prompts help users align settings with privacy goals and reduce risks.

5.2.3. Supporting APFs During In-App Usage. Once users are actively engaging with the app, it’s important to provide ongoing support for anonymity features and ensure their usability.

Support Users in Evaluating Anonymity Effectiveness. Apps should clearly show what data are collected and whether any PII is linked to the account. For example, PA9 believed Cycles collected little PII but still felt uncertain due to limited visibility. A “Your Anonymity Status” section could confirm linked data, display what is stored, and offer

options to manage or remove it. Such visibility reassures users and supports more informed decision-making about their privacy. Prior research notes that users often view PII as accumulating over time, making it critical to offer mechanisms that reflect the temporality of anonymity—that is, how anonymous a user remains as they continue to use the platform. The ability to regularly assess and monitor anonymity is critical for maintaining user trust and a sense of safety, especially in contexts involving sensitive data.

Improve Visibility of Anonymous Opt-In Options.

Anonymity settings should be easy to locate. Participants found features like Flo’s Anonymous Mode difficult to find, as they were buried in account settings, causing frustration and reduced trust. Apps should place these options prominently in main menus and highlight them with distinct visuals. Greater visibility lowers user effort and supports effective anonymity management.

6. Limitation and Future Work

As our participants were predominantly from higher-education backgrounds, our sample may under-represent less-educated users. This does not diminish the significance of our findings, as even highly educated participants encountered difficulties with APFs, highlighting the severity of usability challenges. Moreover, prior work suggests that most PFT app users are relatively well-educated [95], so our sample reflects a key segment of the user population, though it still likely under-represents marginalized communities disproportionately affected by inadequate reproductive privacy, including surveillance, stigma, and discrimination [25]. Future studies should better engage these groups to capture the full spectrum of user vulnerabilities and needs.

Our study audited APF interfaces, highlighting critical usability gaps. Since front-end interfaces and back-end mechanisms play complementary roles in protecting user privacy, future work should evaluate APF implementations at the technical level. By identifying which APFs users notice, rely on, or misunderstand, our findings lay an empirical foundation for targeted backend evaluations that are more feasible and impactful.

7. Conclusion

We conducted a usability study with 32 participants to evaluate APFs in six PFT apps. Using a mixed-methods approach, we identified key challenges such as fragmented user journeys, lack of transparency, and unintuitive privacy designs. From these findings, we propose recommendations for more cohesive anonymity approaches and for APFs that are intuitive, user-friendly, and clearly communicated throughout app use. These insights aim to strengthen APF usability in reproductive health data surveillance and other sensitive contexts.

Acknowledgment

We sincerely thank the anonymous reviewers for their insightful feedback and our study participants for their valuable contributions.

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Appendix A. App Selection Process

This appendix provides additional details on the process used to identify the PFT apps included in the usability testing study. As 3.2 noted, the initial selection yielded 53 apps. We then conducted a detailed review to identify those that explicitly claimed to offer—and demonstrably provided—APFs. We examined five sources: (1) *Roe v. Wade*-related statements, (2) terms of service, (3) privacy policies, (4) app store descriptions, and (5) in-app language. The first author searched for identifiability-related terms (e.g., “anonymous,” “anonymized,” “identify,” “de-identified”) across iOS and Android versions. This approach followed literature framing anonymity as a point on the identifiability spectrum [1], [3].

Apps that only anonymized data before third-party sharing were excluded, since users remained identifiable to the app itself. We focused instead on apps offering features that supported anonymity both externally and within the app’s ecosystem. This yielded seven apps advertising APFs: Stardust, Cycles, Femia, Flo, PD, Natural Cycles, and Femometer. We then reviewed each app to verify its claims, assessing whether users could engage without providing PII (e.g., name, email, social accounts). Between January and February 2024, the first author systematically tested Android and iOS versions across registration, everyday use, and discontinuation, emphasizing how APFs were embedded in privacy settings and controls. During this period, the team held regular meetings to review the screenshots and notes collected by the first author. Although all seven apps advertised privacy features, only six offered functional PII-free options. Natural Cycles, despite announcing an “anonymous

experience” in May 2022 [96], had not implemented it by March 2024 and was excluded from testing.

The six remaining apps—Cycles, Femia, Flo, PD, Femometer, and Stardust—allowed users to track periods and fertility without providing PII such as email, name, phone number, or Apple ID, and thus were selected for the usability testing study. For example, Stardust used third-party authentication to separate PII from usage data, preventing direct access, while Flo’s “Anonymous Mode” enabled transferring health data to a new account stripped of identifiers (e.g., email, name, technical identifiers) [34], [47].

Appendix B. Factor Loadings of the Factors (CFA Results)

Perceived Privacy Risk (AVE = 0.849)		
Item	Statement	Loading
Risk1	It would be risky to give my information to this app.	0.952
Risk2	There would be high potential for privacy invasion associated with giving my information to this app.	0.988
Risk3	There would be too much uncertainty associated with giving my information to this app.	0.908
Risk4	Providing this app with my information would involve many unexpected problems.	0.936
Risk5	I would feel safe giving my information to this app. (reverse-scored item)	-0.813
Perceived Control over Privacy (AVE = 0.841)		
Control1	How much control do you feel you have over the amount of your personal information collected by the app?	0.870
Control2	How much control do you feel you have over who can get access to your personal information in the app?	0.841
Control3	How much control do you feel you have over your personal information that has been released in the app?	0.976
Control4	How much control do you feel you have over how your personal information is being used by the app?	0.915
Control5	Overall, how much in control do you feel you have over your personal information provided to the app?	0.974
Trust (AVE = 0.779)		
Trust1	This app would be trustworthy in handling user information.	0.949
Trust2	This app would tell the truth and fulfill promises related to user information provided by me.	0.860
Trust3	I trust that this app would keep my best interests in mind when dealing with user information.	0.855
Trust4	This app is in general predictable and consistent regarding the usage of user information.	0.864

Appendix C. Correlations Between Factors and the Square Root of AVEs

Factor	Risk	App Control
App Control	-0.735***	—
Trust	-0.756***	0.750***

Note: Each cell is the correlation coefficient between two factors. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All correlation coefficients are smaller than the corresponding square root of AVEs.

TABLE 1. DEMOGRAPHICS AND PARTICIPANTS' PERIOD AND FERTILITY TRACKING EXPERIENCE

P #	Age	Gender	Race/Ethnicity	Education	OS	Tracking Experience	Apps Used	Time (min)
TX1	31-45	Female	Asian	Doctoral	iOS	Currently tracking	MeetYou	73
TX2	18-30	Female	Asian	Doctoral	iOS	Currently tracking	MeetYou	90
TX3	31-45	Female	White	Doctoral	iOS	Never tracked (interested)	N/A	70
TX4	31-45	Female	Asian	Doctoral	iOS	Currently tracking	Apple Health	100
FL1	31-45	Female	Asian	Doctoral	iOS	Currently tracking	MeetYou	140
FL2	18-30	Female	Prefer not	Master's	iOS	Previously tracked	MeetYou	125
FL3	18-30	Female	Asian	Some college	iOS	Currently tracking	Flo/Clue	72
FL4	18-30	Female	Hispanic	Bachelor's	iOS	Currently tracking	Flo/Clue	112
FL5	18-30	Female	White	Master's	iOS	Currently tracking	Apple Health	66
FL6	18-30	Female	Asian	Master's	iOS	Never tracked (interested)	Flo	61
FL7	18-30	Female	Asian	Doctoral	iOS	Currently tracking (others)	N/A	95
FL8	18-30	Female	White	Some college	iOS, Android	Currently tracking	Flo, Eve	90
FL9	18-30	Female	Black/African American	Associate's	iOS	Never tracked (interested)	N/A	66
FL10	18-30	Female	Black/African American	Associate's	iOS, Android	Currently tracking	Ovia	48
PA1	31-45	Female	Asian	Master's	iOS	Currently tracking	Apple Health	106
PA2	18-30	Female	Hispanic	Bachelor's	Android	Previously tracked	Flo/PD/Period calendar	210
PA3	31-45	Female	Asian	Doctoral	iOS, Android	Currently tracking	Period Tracker GP	128
PA4	31-45	Female	White	Bachelor's	iOS	Currently tracking	Flo	125
PA5	31-45	Female	Asian	ABD	Android	Currently tracking	Korean App	157
PA6	31-45	Female	White	Master's	Android	Never tracked (interested)	N/A	137
PA7	31-45	Female	White	Master's	Android	Previously tracked	Flo	102
PA8	18-30	Female	Black/African American	Bachelor's	iOS	Previously tracked	Apple Health	147
PA9	31-45	Female	White	Associate's	iOS	Currently tracking	Apple Health	117
PA10	18-30	Female	Hispanic	In college	iOS	Currently tracking	Fitbit	128
PA11	31-45	Female	Asian	Master's	iOS	Currently tracking	Manual calendar	120
PA12	46-60	Female	White	Doctoral	iOS	Currently tracking	Flo/Period Tracker	87
PA13	18-30	Female	White	Some college	iOS, Android	Currently tracking	Flo/NC/Garmin/Apple Health	110
PA14	31-45	Female	White	Bachelor's	iOS	Currently tracking	PreMom/Easy@home	120
PA15	31-45	Female	Asian	Master's	iOS, Android	Currently tracking	Clue	115
PA16	18-30	Female	Turkish	Master's	iOS, Android	Previously tracked	Clue	159
PA17	31-45	Female	Hispanic	Bachelor's	iOS	Previously tracked	Clue	106
PA18	18-30	Female	White	Bachelor's	iOS, Android	Previously tracked	Period Tracker GP	112

Appendix D. Demographics and Details Regarding Participants' Experience With Period and Fertility Tracking

Appendix E. Usability Testing Study Task Design

This appendix provides additional details on app interaction tasks during the usability testing. We asked participants to complete specific task(s) on three apps on a testing phone provided by the researchers.

Pre-Task Setup. The screener survey gathered information on participants' familiarity with mobile operating systems(OS) and their prior use of PFT apps. Each participant received a phone with their preferred OS and was assigned three unfamiliar apps from a pool of six to avoid familiarity bias. App order was counterbalanced to reduce learning effects.

Task procedures. At the beginning of the task session, we first inform the participants that they could stop the task at any point if they were unable to complete it. Then we introduce two tasks mentioned in 3.2. As 3.2 noted, Task 1

mirrors the goal of preventing PII disclosure upfront, while Task 2 aligns with the need for users to remove PII post-registration. Not all apps supported both tasks. Flo, with features like "Anonymous Mode," allowed participants to complete both. In contrast, Period Diary, Cycles, Femometer, Femia, and Stardust lacked post-registration unlinking options, so participants using these apps completed only Task 1.

Participants were instructed to think aloud while interacting with the apps, sharing their thoughts, confusions, and suggestions. If they fell silent, researchers gently prompted them. Tasks were untimed, and participants could explore additional privacy features if desired. Researchers observed without offering help and introduced the next task only when participants chose to proceed.

Appendix F. Additional Study Materials

We have made additional study materials (i.e., user flow of completing tasks on each app, usability study protocol and surveys) publicly available on the Open Science Framework: https://osf.io/5tgmu/?view_only=9e71c8e2ee3a4305a4ba5f48193cc103.

Appendix G. Meta-Review

The following meta-review was prepared by the program committee for the 2026 IEEE Symposium on Security and Privacy (S&P) as part of the review process as detailed in the call for papers.

G.1. Summary

This paper presents a mixed-method investigation into usability, and perception thereof, of anonymity features in reproductive health care apps. In the motivation, the paper connects this to the serious legal threats to live and health people are exposed to when simply using healthcare relevant applications.

The paper performs a mixed methods study where ca. 32 participants receive tasks to utilize anonymity and privacy preserving features (disabling of ad tracking/sharing of private information with third parties) of various reproductive healthcare apps.

The paper finds serious limitations in the observed apps, and provides recommendations for better user flows.

G.2. Scientific Contributions

- Addresses a Long-Known Issue
- Provides a Valuable Step Forward in an Established Field

G.3. Reasons for Acceptance

- 1) The paper handles a serious topic, where the implications of tracking and privacy violations so far were-based on the paper's findings regarding the prevalence of ad tracking and third-party sharing of information—not in the focus of providers of reproductive health care apps.
- 2) Current global events provide additional urgency to sharing the results with the community. Especially the perspective on choice within the paper, i.e., usability of executing choice in terms of privacy holds notable opportunity for sparking wider discussions in the community on whether choice should be required in specific application cases and health critical applications, or privacy and anonymity should just be the default.

G.4. Noteworthy Concerns

- 1) Given the reviewers positionality, it should be noted that they may have a different connection to the subject matter than major parts of the community. While confident that their positionality did not influence their assessment, the impact of this diverging connection to the subject may not necessarily be intrinsically apparent. As such, the reviewers appreciate readers' being cognizant of this when interpreting the paper.

Appendix H. Response to the Meta-Review

We greatly appreciate the meta-review's effort to provide additional context about the reviewers' backgrounds and perspectives. We agree that, while this note is not about our paper itself, it is of great value for reviewers' positionality regarding the subject matter to be transparent to readers, as this helps them remain aware of potential unconscious biases and contextual influences in the review process. We appreciate this transparency and believe it will be helpful for readers when interpreting the review.